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On the Born series methods for solving inhomogeneous Helmholtz equation in biomedical photoacoustics

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Abstract

The Born series methods, namely, traditional Born series (TBS) and convergent Born series (CBS), have been recently implemented to numerically solve the time-independent photoacoustic (PA) wave equation for an acoustically inhomogeneous source. The TBS algorithm diverges when the sound-speed mismatch is >20%, however, the CBS technique provides faithful result even beyond this limit. These protocols are iterative in nature and computationally expensive. Currently, MATLAB-based implementations are available to study PA emission from a single source mimicking a cell. However, efficient numerical implementation strategy is further needed particularly to calculate PA field from a tissue. Therefore, to develop insights, uniprocessor-based C codes were realized for these schemes. The PA field (in 2D) was computed at a distance 35 μm from a source (a light absorbing disc of radius 7.5 μm) over a frequency range from 7.32 to 512 MHz with a resolution of 7.32 MHz. The sound-speed within the source region was varied from v_s = 1200, 1500 and 1800 m/s, but the same quantity for the ambient medium was fixed to v_f = 1500 m/s. The C program was found to be at least 10 times slower than the corresponding MATLAB program. It may be because MATLAB inherently implements parallel computing while evaluating the forward and backward fast Fourier transforms (FFTs) associated with the Born series approaches. Multiprocessor-based FFTs and parallel nested loops are being incorporated into the C program for enhancement of its execution speed.

Keywords: Optoacoustics, Photoacoustics, Inhomogeneous wave equation, Traditional Born series, Convergent Born series.

Introduction

The solution to the inhomogeneous Helmholtz wave equation is required in many branches of engineering, science and medicine etc. [1, 2]. For simple geometries, exact analytical solutions can readily be derived (e.g., layer, infinite cylinder, sphere etc.) [3]. The solution is deduced by matching the wave function and its derivative inside and outside the source. Several numerical techniques are also available to solve such an equation. For example, pseudo spectral method, finite element method can be employed.

The solution may also be obtained by evaluating the integral equation involving the Green's function technique. The solution in this method essentially appears as an infinite series. Such a series sum is known as the traditional Born series (TBS). The TBS scheme provides faithful solution when the size and strength of the scattering potential are small but the solution does not converge when these conditions do not met. Recently, Osnabrugge et al. have developed a method called the convergent Born Series (CBS) introducing a preconditioner into the TBS expression [4]. The CBS protocol works well for arbitrary large media with strong scattering potential. The TBS and CBS methods have recently been implemented to solve the inhomogeneous photoacoustic (PA) wave equation [5, 6, 7]. There exist two source terms in this equation. The first term is responsible for conversion of optical energy into acoustical energy and the second term arises due to the mismatch of speed of sound between the source region and the ambient medium. These protocols are iterative in nature and computationally expensive.

In the previous works, the TBS and CBS algorithms were implemented in MATLAB [5, 6, 7]. It could handle a large computational domain of size 2148 × 2148 grid points with a spatial resolution of $dx = dy$ = 100 nm. However, this computational environment may not be suitable for tissue simulation (i.e., collection of cells) viz-a-viz to study spectral properties of PA emission from tissue. With this motivation, we recently developed uniprocessor-based C programs realizing the TBS and CBS methods. In this work, we compare computational speeds of these approaches. We found that MATLAB programs are still faster than the C codes. The implementation steps, comparison of execution speeds and physical reasons why the MATLAB code runs faster than the C code are presented elaborately in the following sections.

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Theoretical aspect

The Helmholtz equation inside and outside the PA source can be written as,

$$\nabla^2\psi(\mathbf{r}) + k_s^2\psi(\mathbf{r}) = \frac{i\omega\mu\beta I_0}{C_P}, \quad \nabla^2\psi(\mathbf{r}) + k_f^2\psi(\mathbf{r}) = 0, \tag{1}$$

respectively. Here, μ, β and C_p are the optical absorption coefficient, isobaric thermal expansion coefficient and specific heat for the absorbing region, respectively; ω are I_0 are the modulation frequency and intensity of laser radiation, respectively; k_s and k_f are the wave numbers inside and outside the PA source, respectively. The expression of the exact solution to Eq. (1) for a solid circular source can be found in the literature [5, 6]. A schematic diagram is given in Fig. 1.

The PA wave equation as written in Eq. (1) can be cast in a compact form as,

$$\nabla^2\psi(\mathbf{r}) + (k_f^2 + i\epsilon)\psi(\mathbf{r}) = -S(\mathbf{r}) - V(\mathbf{r})\psi(\mathbf{r}), \tag{2}$$

where ϵ is an infinitesimally small real number and $S(\mathbf{r})$ and $V(\mathbf{r})$ being the source term and the scattering potential, respectively; these are defined as, $S(\mathbf{r}) = -i\mu\beta I_0\omega/C_p$ inside the source else 0; $V(\mathbf{r}) = k_s^2 - k_f^2 - i\epsilon$ inside the source else $-i\epsilon$.

The solution of Eq. (2) in the TBS and CBS methods can be obtained as,

$$\psi_{TBS} = [1 + GV + GVGV + ...]GS \quad \text{and} \quad \psi_{CBS} = [1 + M + M^2 + ...]\gamma GS, \tag{3}$$

respectively, where $\gamma = iV(\mathbf{r})/\epsilon$, $\epsilon \geq \max|k_s^2 - k_f^2|^2$ and $M = \gamma GV - \gamma + 1$. The notation G is used to indicate Green's function. In the frequency domain, it takes the form, $\tilde{g}(\mathbf{p}) = 1/(|\mathbf{p}|^2 - k_f^2 - i\epsilon)$. Moreover, G and $\tilde{g}(\mathbf{p})$ are related as $G = F^{-1}\tilde{g}(\mathbf{p})F$ where F and F^{-1} are the forward and inverse Fourier transform operators. The infinite series sums of Eq. (3) are numerically implemented in this work for a solid circular source.

Simulation methodology

The size of the computational domain was 2048 × 2048 grid points with separation $dx = dy = 100$ nm. An absorbing layer of 100 grid points was placed at each boundary as shown in Fig.1. The density inside and outside the source was taken as the same ($\rho_s = \rho_f = 1000$ kg/m^3); ϵ was fixed to $0.8k_f^2$. Other optical and thermo-mechanical parameters were fixed to unity based on the literature [5, 6, 7]. The speed of sound of the ambient medium was fixed at $v_f = 1500$ m/s. The speed of sound within the source was varied as $v_s = 1800$, 1500 and 1200 m/s. A solid homogeneous circular region ($a = 7.5$ μm) was considered as the PA source (see Fig. 1). A point detector was positioned at a distance of 35 μm from the center to record pressure data. The PA fields were computed at frequencies from 7.3 to 512.78 MHz with an increment of 7.3 MHz for the exact, TBS and CBS methods. The pressure data for the TBS and CBS algorithms were computed in the following manner,

$$\psi_{TBS_{n+1}}(\mathbf{r}) = \text{ifft}_2[\tilde{g}(\mathbf{p})\text{fft}_2[V(\mathbf{r})\psi_{TBS_n}(\mathbf{r}) + S(\mathbf{r})]], \tag{4}$$

and

$$\psi_{CBS_{n+1}}(\mathbf{r}) = \psi_{CBS_n}(\mathbf{r}) - (i/\epsilon)V(\mathbf{r})(\psi_{CBS_n}(\mathbf{r}) - \text{ifft}_2[\tilde{g}(\mathbf{p})\text{fft}_2[V(\mathbf{r})\psi_{CBS_n}(\mathbf{r}) + S(\mathbf{r})]]), \tag{5}$$

respectively; fft2 and ifft2 stand for the forward and inverse fast Fourier transforms in 2D, respectively. For $n = 0$, initial pressure fields in TBS and CBS methods were taken as,

$$\psi_{TBS_0}(\mathbf{r}) = \text{ifft}_2[\tilde{g}(\mathbf{p})\text{fft}_2 S(\mathbf{r})] \quad \text{and} \quad \psi_{CBS_0}(\mathbf{r}) = \gamma(\text{ifft}_2[\tilde{g}(\mathbf{p})\text{fft}_2 S(\mathbf{r})]), \tag{6}$$

respectively. The iterative computations were continued until the PA fields converged in the two methods.

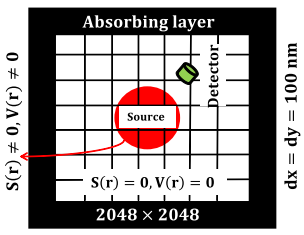


Figure 1: Computational domain.

Methods	v_s in m/s	Time in minutes
TBS (MATLAB)	1200, 1500, 1800	106, 22.8, 19.9
CBS (MATLAB)	1200, 1500, 1800	31.5, 39.7, 34
TBS (C-Programming)	1200, 1500, 1800	1440, 320, 325
CBS (C-Programming)	1200, 1500, 1800	359, 400, 394

Table 1: Execution time required for convergence.

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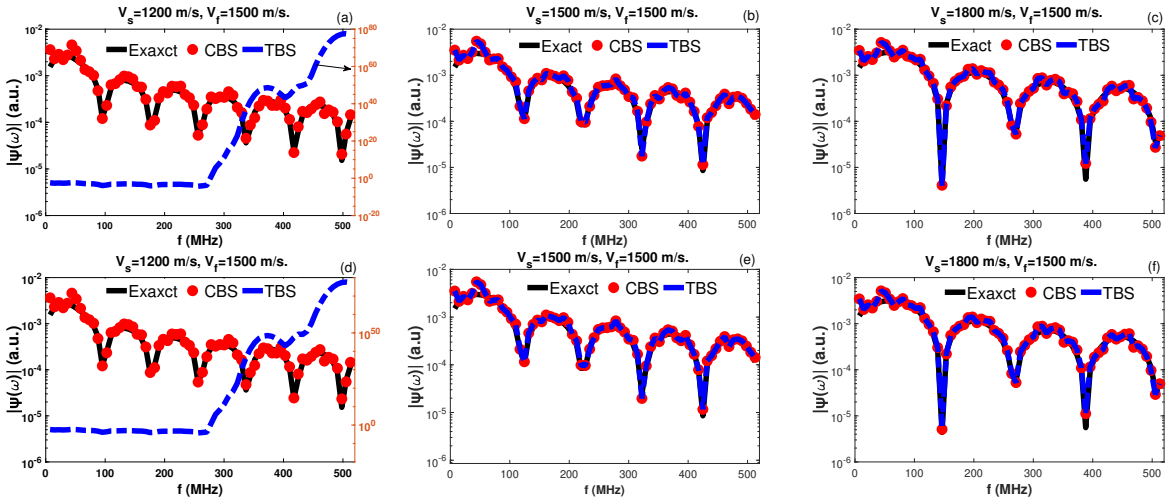


Figure 2: The variation of $|\psi|$ as a function of frequency for the exact/analytical, TBS and CBS procedures. Top row presents results for the MATLAB programs whereas the bottom row displays those of the C codes.

Numerical results

The numerical results are summarized in Fig. 2. Both the TBS and CBS methods showed good agreement with the exact result over the entire frequency range for a circular source. In case of $v_s = 1200$ m/s, the TBS method diverges. The execution time consumed by various programs are written in Table 1. It can be observed that a MATLAB code takes at least 10 times less time than the C counterpart in the same computer [5, 6, 7].

Discussion and conclusions

The TBS method is preferable when the mismatch in speed of sound is small (approximately < 10%) because it takes less time to compute the PA field compared to the CBS technique. However, for large mismatch, the CBS method is always preferable. Our simulation results suggest that uniprocessor-based C codes take longer time to execute than the MATLAB codes. It may be because MATLAB inherently implements multiprocessor-based computation while evaluating the forward and backward fast Fourier transforms (FFTs) associated with the Born series approaches. In order to increase computational speed of the C program, currently multiprocessor-based FFTs and parallelization of the nested loops are being carried out.

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